

FSUV. 2019. Spring. #8.

Let $f: (0,1) \rightarrow \mathbb{R}$ be a convex map.

1). For any $0 < x < y < 1$, $f'_+(x) \leq f'_-(y)$.

2). The set $\{x \in (0,1) \mid f \text{ is not differentiable at } x\}$ is at most countable.

Proof. First, since f is convex, for any $x_1 < x_2 < x_3$ in $(0,1)$,

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_3) - f(x_1)}{x_3 - x_1} \leq \frac{f(x_3) - f(x_2)}{x_3 - x_2}$$

1). $f'_+(x)$ exists for any $x \in (0,1)$

Let $x \in (0,1)$ be given. And, let $a < x$ be fixed.

Then, for any $0 < h$,

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(x+h) - f(x)}{h}, \text{ and since } \frac{f(x+h) - f(x)}{h} \text{ decreases as } h \rightarrow 0,$$

so $f'_+(x)$ exists. Similarly $f'_-(x)$ exists.

2). $x < y \Rightarrow f'_+(x) \leq f'_-(y)$.

Take $\delta = \frac{1}{2}(x-y)$. Then, for all $h \in \mathbb{R}$ with $0 < h < \delta$,

$$\frac{f(x+h) - f(x)}{x+h-x} < \frac{f(y) - f(y-h)}{y-(y-h)},$$

So the first claim proved.

Now, let $x \in X$ be a point such that f is not differentiable.

Since $f'_+(x)$ and $f'_-(x)$ exist, the only possibility is

$$f'_-(x) < f'_+(x).$$

By the density of rationals, there is $r \in (f'_-(x), f'_+(x)) \cap \mathbb{Q}$.

And, for any $x < y$, $f'_+(x) \leq f'_-(y)$, if y is another not differentiable point, $x < y$,

then $r_x < r_y$, so the map from the set into $\mathbb{Q}$; $x \mapsto r_x$ is injective.

So, the set is at most countable.

FSNU. 2019, spring, #14,

Let $R = \mathbb{Z}[\sqrt{-5}]$. Prove or disprove

a) $1 \pm \sqrt{-5}$ are irreducible

b) $1 \pm \sqrt{-5}$ are prime.

Proof. If $1 + \sqrt{-5}$ is reducible, then $1 + \sqrt{-5} = \alpha\beta$ for some non-unit α, β .

Now, the norm $N: R \rightarrow \mathbb{N}$ is defined by $N(a + b\sqrt{-5}) = a^2 + 5b^2$.

$$N(1 + \sqrt{-5}) = 6 = N(\alpha)N(\beta)$$

If $N(\alpha) = 1$ or $N(\beta) = 1$, then $\alpha = \pm 1$ or $\beta = \pm 1$, which is a contradiction, because

$$N(\alpha) = 1 \Leftrightarrow \alpha = a + b\sqrt{-5} \text{ satisfies } a^2 + 5b^2 = 1 \Leftrightarrow b = 0, a = \pm 1.$$

So neither $N(\alpha)$ nor $N(\beta)$ is 1.

Now, suppose $N(\alpha) = 2$. Then, $\alpha = a + b\sqrt{-5}$ satisfies $a^2 + 5b^2 = 2$,

but it cannot have an integer solution and similarly if $N(\alpha) = 3$.

Similarly $1 - \sqrt{-5}$ is not irreducible.

$$\text{And, observe that } (1 + \sqrt{-5})(1 - \sqrt{-5}) = 6 = 2 \cdot 3.$$

Therefore, if $1 + \sqrt{-5}$ is prime, it divides either 2 or 3.

$$\text{But, } N(2) = N(1 + \sqrt{-5}) \cdot N(\alpha)$$

$$\Rightarrow 4 = 6 \cdot N(\alpha), \text{ there is no such } \alpha \in R.$$

And similarly $N(3) = 9 = 6 \cdot N(\alpha)$, there is no such $\alpha \in R$ so it is not prime.

Similarly $1 - \sqrt{-5}$ is not prime.

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Find a fraction Ring of $\mathbb{Z}[2i]$, and prove that it is not U.F.D.

Prbf. Since $\mathbb{Z}[2i] = \{a + 2bi \mid a, b \in \mathbb{Z}\}$,

And the inverse of $a + 2bi$ is

$$\frac{a - 2bi}{a^2 + 4b^2} \quad \text{if } (a, b) \neq (0, 0)$$

So, the fraction Ring is

$$\mathbb{Q}^{-1} \mathbb{Z}[2i] = \left\{ \frac{a - 2bi}{a^2 + 4b^2} \cdot (c + 2di) \mid \begin{array}{l} a, b, c, d \in \mathbb{Z}, \\ (a, b) \neq (0, 0) \end{array} \right\}$$

$$(\mathbb{Z}[2i] \cong \mathbb{Z}[x]/(x^2 + 2) \cong \mathbb{Q}(i))$$

And, observe that

$$(2 + 2i)(2 - 2i) = 8 = 2 \cdot 2 \cdot 2$$

But, $2 \pm 2i$ are irreducible, because: if $2 + 2i = \alpha\beta$, then

$N(\alpha)N(\beta) = 8$, so $N(\alpha) = 2$ or 4 (if $=1$, then it is unit),

$N(\alpha) = 2$, then $\alpha = a + 2bi$ satisfies $a^2 + 4b^2 = 2$, it cannot have integer solution,

$N(\alpha) = 4$, then $N(\beta) = 2$, but similarly impossible.

And 2 is irreducible, similarly, $2 = \alpha\beta$ then $N(\alpha)N(\beta) = 4$, so $N(\alpha) = 2$, impossible.

Now, 8 has different two irreducible factorizations, contradiction.

TSU. 2019 Fall, #1.

ISNU 2019 Spring #17.

Let $f(z) = a_0 + a_1 z + a_2 z^2 + \dots + a_{d-1} z^{d-1} + a_d z^d \in \mathbb{C}[z]$, $a_d \neq 0$.

Evaluate $\lim_{R \rightarrow \infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz$.

Proof. $\log f(z) = \log(z-r_1)(z-r_2) \cdots (z-r_d) = \sum \log \frac{1}{z-r_i}$.

$$\text{so, } \frac{f'(z)}{f(z)} = \sum \frac{1}{z-r_i}, \quad z \cdot \frac{f'(z)}{f(z)} = \sum \frac{z}{z-r_i}.$$

Now, since $\text{Res}(r_i) = \lim_{z \rightarrow r_i} (z-r_i) \cdot \frac{z}{z-r_i} = r_i$, (or $\frac{z}{z-r_i} = 1 + \frac{r_i}{z-r_i}$)

$$\text{There for, } \int_{|z|=R} \frac{z f'(z)}{f(z)} dz = 2\pi i \sum \text{Res}(r_i) = 2\pi i \cdot \sum (r_i) =$$

[SNU 2021 Spring #8, Robbins-Siegmund Lemma.

Let $\{V_k\}, \{S_k\}, \{U_k\}$ be non-negative real sequences satisfying

$$V_{k+1} \leq V_k - S_k + U_k \text{ for all } k \in \mathbb{N} \text{ and } \sum_{k=0}^{\infty} U_k < \infty$$

Then, $S_k \rightarrow 0$ and $\{V_k\}$ has a limit as $k \rightarrow \infty$.

Proof. observe that

$$V_{k+1} - V_k + S_k \leq U_k \text{ for all } k \in \mathbb{N}.$$

Therefore, given $N > 1$,

$$\sum_{k=1}^N (V_{k+1} - V_k + S_k) = V_{N+1} - V_1 + \sum_{k=1}^N S_k \leq \sum_{k=1}^N U_k \leq \sum_{k=1}^{\infty} U_k$$

thus
$$\sum_{k=1}^N S_k \leq \sum_{k=1}^{\infty} U_k + V_1 - V_{N+1} \leq \sum_{k=1}^{\infty} U_k + V_1 < \infty,$$

So $\sum_{k=1}^{\infty} S_k$ converges, $S_k \rightarrow 0$ as $k \rightarrow \infty$.]

Meanwhile, define

$$W_k = V_k - \sum_{i=1}^{k-1} U_i \quad (k \geq 2).$$

Then, for any $k \geq 2$,

$$W_{k+1} - W_k = V_{k+1} - V_k - U_k \leq -S_k \leq 0, \text{ so } \{W_k\} \text{ is monotone decreasing.}$$

Moreover, given $k \geq 2$,

$$W_k = V_k - \sum_{i=1}^{k-1} U_i \geq -\sum_{i=1}^{k-1} U_i \geq -\sum_{i=1}^{\infty} U_i$$

thus, W_k is bounded below, so converges. Now,

$$V_k = \sum_{i=1}^{k-1} U_i + W_k$$

Converges, being sum of convergent sequences.

FSM 2022 Spring #8.

Define $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$

is differentiable.

Proof. Since $\sum_{n=1}^N \frac{1}{x^2+n^2} \leq \sum_{n=1}^N \frac{1}{n^2} < \sum_{n=1}^{\infty} \frac{1}{n^2}$ for any $x \in \mathbb{R}$,

the Weierstrass M-test gives the series converges absolutely,

Moreover, $\frac{d}{dx} \left(\frac{1}{x^2+n^2} \right) = -\frac{2x}{(x^2+n^2)^2}$

Satisfies: for $|x| \geq 1$,

$$\left| \frac{x}{(x^2+n^2)^2} \right| \leq \left| \frac{x^2}{(x^2+n^2)^2} \right| \leq \left| \frac{x^2+n^2}{(x^2+n^2)^2} \right| = \frac{1}{x^2+n^2} \leq \frac{1}{n^2}$$

and for $|x| < 1$,

$$\left| \frac{x}{(x^2+n^2)^2} \right| \leq \frac{1}{(x^2+n^2)^2} \leq \frac{1}{n^4}.$$

Now, Weierstrass M-test again, $\frac{d}{dx} \left(\sum_{n=1}^N \frac{1}{x^2+n^2} \right)$ converges uniformly on $\mathbb{R}$.

Now, f is differentiable, with $f'(x) = -\sum_{n=1}^{\infty} \frac{2x}{(x^2+n^2)^2}$.

ISNU 2018, Fall, #10.

Let $\{a_n\}$ be a non-negative real numbers satisfying

$$a_{n+m} \leq a_n + a_m \text{ for } n, m \geq 1.$$

Show that $\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \in \mathbb{N}} \left\{ \frac{a_n}{n} \right\}$. $\sqrt{5} < \sqrt{2} \sqrt{3}$
 $2^2 + 3^2 = 13$
 $2 \cdot 3 = 2 \cdot 5$

Proof. Given $k > 1$,

$$a_k = a_{k-1+1} \leq a_{k-1} + a_1 \leq a_{k-2} + 2a_1 \leq \dots \leq k \cdot a_1$$

If $a_1 = 0$, then $\{a_n\}$ is just zero. Assume $a_1 > 0$.

For any $n \in \mathbb{N}$,

$$\frac{a_n}{n} \leq \frac{n \cdot a_1}{n} = a_1 \quad \text{and} \quad \frac{a_{n+1}}{n+1} \leq \frac{a_n + a_1}{n+1} \leq \frac{a_n + a_1}{n}$$

Given $\epsilon > 0$, there is $N \in \mathbb{N}$ such that

$$\inf \frac{a_n}{n} \leq \frac{a_N}{N} < \inf \frac{a_n}{n} + \epsilon.$$

$m > n$, then

$$\frac{a_m}{m} - \frac{a_n}{n} \leq$$

$$\frac{a_{k+n}}{k+n} - \frac{a_n}{n}$$

ISNU. 2025 Fall #5.

Let G be an order of 21 subgroup of $\mathbb{C}^\times$.

a). G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions.

Proof By the Sylar Theorem, there exist $P \in \text{Syl}_3(G)$ and $Q \in \text{Syl}_7(G)$.

Perote $P = \langle x \rangle$ and $Q = \langle y \rangle$ for some $x \in P$ and $y \in Q$,

since both P and Q have Prime order, so cyclic.

Moreover, since $\mathbb{C}^\times$ is abelian, x and y are commutative, so

$|xy| = 21$, so $G = \langle xy \rangle$, cyclic.

To show the second assertion, since G is cyclic, we can write

$G = \langle s \rangle$. Given $g \in G$, $g = s^a$, for some $0 \leq a \leq 21$.

So, for some $x = s^b$,

$$x^{12} = g^{15} \Leftrightarrow s^{12b} = s^{15a} \Leftrightarrow 12b \equiv 15a \pmod{21}$$

$$\Leftrightarrow \gcd(12, 21) \mid 15a.$$

Since $0 \leq b \leq 21$, the number of solutions is $\gcd(12, 21) = 3$.

[Note: $ax \equiv b \pmod{n}$ has a solution for $x \Leftrightarrow (a, n) \mid b$

SNV. 2020 Spring 4.

Let R be a commutative ring, and $I \neq R$ is a prime ideal.

Then, the radical of I

$$\sqrt{I} = \{a \in R \mid a^m \in I \text{ for some } m \in \mathbb{N}\} \text{ is prime ideal.}$$

Proof. Actually, $I = \sqrt{I}$.

Given $a \in I$, $a^1 \in I$, so $a \in \sqrt{I}$. Thus $I \subseteq \sqrt{I}$.

Conversely, let $a \in \sqrt{I}$ be given. Then, $a^m \in I$ for some $m \in \mathbb{N}$.

Since I is prime, either a or $a^{m-1} \in I$. If $a \in I$, then we are done.

If $a^{m-1} \in I$, again, either a or $a^{m-2} \in I$.

Inductively, we obtain a must be contained in I , so proved.

SNV. 2020 Spring #5.

Find the Galois group of $x^5 - 3 \in \mathbb{Q}(\zeta)[x]$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive fifth root of unity.

Proof. Let K be the splitting field of $x^5 - 3$.

Since all roots of $x^5 - 3$ are

$$\sqrt[5]{3} \cdot \zeta^k \quad (k=0,1,2,3,4) \text{ where } \zeta = \exp\left(2\pi i \cdot \frac{1}{5}\right) = e^{\frac{2\pi i}{5}},$$

Therefore $K = \mathbb{Q}(\sqrt[5]{3}, \zeta)$. Now,

$$K = \mathbb{Q}(\sqrt[5]{3}, \zeta)$$

$$\begin{array}{cc} 4 & \backslash & 5 \\ \mathbb{Q}(\sqrt[5]{3}) & & \mathbb{Q}(\zeta) \end{array}$$

$$\begin{array}{cc} 5 & \backslash & 4 \\ \mathbb{Q} & & \mathbb{Q} \end{array} \quad \left(\begin{array}{l} x^4 + x^3 + x^2 + x + 1 \text{ has root } \zeta \\ \text{and 'irreducible'} \end{array} \right)$$

($\because x^5 - 3$ is 'irreducible over $\mathbb{Q}$)

Since $[\mathbb{Q}(\sqrt[5]{3}, \zeta) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}] \cdot [\mathbb{Q}(\zeta) : \mathbb{Q}]$ and both degree over $\mathbb{Q}$ are relatively prime,

$[K : \mathbb{Q}] = 20$. And, since $[K : \mathbb{Q}(\zeta)] = 5$, prime,

$$\text{Gal}(K/\mathbb{Q}(\zeta)) \cong \mathbb{Z}_5.$$

[SNU. 2020 Spring, #9,

Suppose that f is an entire and $\operatorname{Re} f(z) \neq 0$ for all $z \in \mathbb{C}$.

Show that f is constant.

Proof Let $h(z) = e^{-f(z)}$. Since $|f(z)| \geq \operatorname{Re} f(z) \neq 0$, the h is well-defined.

$$\text{And, } |h(z)| = \left| \frac{1}{e^{\operatorname{Re} f(z) + i \operatorname{Im} f(z)}} \right| = \frac{1}{|e^{\operatorname{Re} f(z)}|} \cdot 1$$

Meanwhile, since f is entire, $\operatorname{Re} f$ is continuous, so $\operatorname{Re} f > 0$ or $\operatorname{Re} f < 0$.